

9.1 Introduction

From the various expressions of the mutual information $I(X;Y)$ in the previous lecture, we can establish interesting relations between joint entropies, conditional entropies and marginal entropies. For example, since $I(X;Y) = H(Y) - H(Y|X) = H(X) + H(Y) - H(X,Y) = H(X) - H(X|Y)$, we can isolate $H(X,Y)$ from the previous equalities combining the second and the third terms, or the third and the fourth terms, to obtain that

$$H(X,Y) = H(Y|X) + H(X) \quad (9.1)$$

$$= H(X|Y) + H(Y), \quad (9.2)$$

the so-called **chain rule of entropy**. Equations (9.1) and (9.2) are a consequence of the chain rule of probability. That is, using that $P_{XY}(x,y) = P_{Y|X}(y|x)P_X(x)$ and that the joint entropy $H(X,Y)$ can be written as the expected value of $-\log_2 P_{XY}(X,Y)$, we have that

$$H(X,Y) = \mathbb{E}[-\log_2 P_{XY}(X,Y)] \quad (9.3)$$

$$= \mathbb{E}[-\log_2 P_{Y|X}(Y|X) - \log_2 P_X(X)] \quad (9.4)$$

$$= \underbrace{\mathbb{E}[-\log_2 P_{Y|X}(Y|X)]}_{H(Y|X)} + \underbrace{\mathbb{E}[-\log_2 P_X(X)]}_{H(X)}. \quad (9.5)$$

Equation (9.2) can be obtained in a similar way using $P_{XY}(x,y) = P_{X|Y}(x|y)P_Y(y)$. Entropy can be interpreted as a measure of uncertainty of a random variable. Hence, the chain rule of entropy tells us that the uncertainty of a pair of random variables $H(X,Y)$ can be decomposed as the uncertainty present in one random variable, for example $H(X)$, and the uncertainty of the other random variable knowing the first one, $H(Y|X)$. The mutual information $I(X;Y)$ has an analog meaning as the information that one random variable has about the other and that this information can be measured as the uncertainty of a random variable $H(Y)$ after removing the uncertainty of this random variable knowing the other $H(Y|X)$. In this last lecture, we will see several relations and properties of information measures (entropies and mutual informations), mathematical tools used in other disciplines beyond digital communications.

9.2 Information measures

Since we can express the joint probability distribution of n random variables $X_1 \dots X_n$ as

$$P_{X_1 \dots X_n}(x_1, \dots, x_n) = P_{X_1}(x_1)P_{X_2|X_1}(x_2|x_1)P_{X_3|X_1X_2}(x_3|x_1x_2) \dots P_{X_n|X_1 \dots X_{n-1}}(x_n|x_1 \dots x_{n-1}) \quad (9.6)$$

we can write the **chain rule of entropy** in general as

$$H(X_1 \dots X_n) = H(X_1) + H(X_2|X_1) + H(X_3|X_1X_2) + \dots + H(X_n|X_1 \dots X_{n-1}). \quad (9.7)$$

Remember that we can write the chain rule in our favorite order. For example, $H(X,Y,Z) = H(X) + H(Y|X) + H(Z|XY)$ but also $H(X,Y,Z) = H(Y) + H(Z|Y) + H(X|YZ)$. Since conditioning reduces entropy, we have that $H(X_2|X_1) \leq H(X_2)$, that $H(X_3|X_1X_2) \leq H(X_3)$ and so on, concluding that for jointly distributed random variables $X_1, \dots, X_n$, the **joint entropy of a sequence** of random variables is upper bounded by the sum of marginal entropies:

$$H(X_1 \dots X_n) \leq \sum_{i=1}^n H(X_i). \quad (9.8)$$

The bound in (9.8) is achieved with equality when $H(X_2|X_1) = H(X_2)$, $H(X_3|X_1X_2) = H(X_3)$ and so on. We know that $H(X_2|X_1) = H(X_2)$ is achieved when $P_{X_2|X_1}(x_2|x_1) = P_{X_2}(x_2)$, that is, when X_1 and X_2 are independent, hence knowing X_1 does not reduce uncertainty in X_2 . Similarly, $H(X_3|X_1X_2) = H(X_3)$ is



Figure 9.1: Output Y as a deterministic function of X .

achieved when X_3 is independent from X_1 and X_2 , and so on. We conclude that (9.8) is satisfied with equality when $(X_1, \dots, X_n)$ is a sequence of independent random variables.

The next result shows that applying a deterministic function to a random variable cannot increase its entropy (see Figure 9.1). Let X and Y be two random variables taking on values on alphabets $\mathcal{X}, \mathcal{Y}$, respectively, and suppose that Y is the output of a deterministic function of X , that is, $Y = g(X)$. A function $g(x)$ is deterministic function if it does not introduce randomness, for example $g(x) = x^2$ or $g(x) = \log_2 p(x)$, since for a given x , $g(x)$ always gives the same result with probability 1. We can then expand the joint entropy between X and Y in the two possible ways as we saw at the beginning of this lecture:

$$H(X, Y) = H(X) + \overbrace{H(Y|X)}^{=0} \quad (9.9)$$

$$= H(Y) + \underbrace{H(X|Y)}_{\geq 0} \quad (9.10)$$

where $H(Y|X) = 0$ since Y is a deterministic function of X and there is no additional uncertainty in Y given X . Since $H(X|Y) \geq 0$, then we have that the entropy of the output of a **deterministic function** is upper bounded by the entropy of the input

$$H(Y) \leq H(X). \quad (9.11)$$

In other words, manipulating X in a deterministic way cannot increase uncertainty.

So far, we have studied multiple properties of entropies. We now move on to define $I(X; Y|Z)$, that is the information that a random variable X has about another one Y knowing a third random variable Z . Using the definition of mutual information from the previous lecture, this quantity called the **conditional mutual information** can be written in multiple ways as

$$I(X; Y|Z) = \mathbb{E} \left[\log_2 \frac{P_{XY|Z}(X, Y|Z)}{P_{X|Z}(X|Z)P_{Y|Z}(Y|Z)} \right] = \mathbb{E} \left[\log_2 \frac{P_{X|YZ}(X|YZ)}{P_{X|Z}(X|Z)} \right] = \mathbb{E} \left[\log_2 \frac{P_{Y|XZ}(Y|XZ)}{P_{Y|Z}(Y|Z)} \right], \quad (9.12)$$

and expanding the logarithms in terms of entropies as

$$I(X; Y|Z) = H(Y|Z) - H(Y|XZ) \quad (9.13)$$

$$= H(X|Z) + H(Y|Z) - H(X, Y|Z) \quad (9.14)$$

$$= H(X|Z) - H(X|YZ) \quad (9.15)$$

$$= I(Y; X|Z). \quad (9.16)$$

These are exactly the same expressions as the ones in lecture 8 after adding the conditioning on Z in all the terms, and are valid for any triplet of random variables with joint distribution $P_{XYZ}(x, y, z)$.

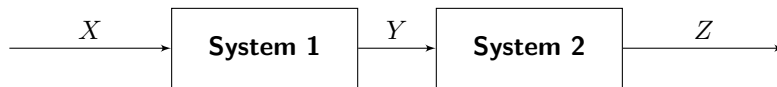


Figure 9.2: Illustration of conditional independence and the data processing theorem.

In digital communications and other disciplines, the random variables X , Y and Z are part of a data processing chain as in Figure 9.2, in the sense that Y is the output of a system whose input is X , and that Z is the output of another system whose input is Y . This does not imply that Z is independent from Y , since a variation on X may induce a variation on Y that, in its turn, may induce a variation in Z ! However, knowing the variable in the middle of the chain, that is Y , indeed produces a kind of independence between the input X and the output Z of the chain. This is the notion of **Markov chain**: three random variables X , Y and Z form a Markov chain $X \rightarrow Y \rightarrow Z$ if given Y , X and Z are independent, that is

$$P_{XZ|Y}(x, z|y) = P_{X|Y}(x|y)P_{Z|Y}(z|y). \quad (9.17)$$

Consequently, if $X \rightarrow Y \rightarrow Z$ form a Markov chain, after knowing Y , Z has no information about X , and vice versa, and the conditional mutual information $I(X; Z|Y)$ vanishes because

$$I(X; Z|Y) = \mathbb{E} \left[\log_2 \frac{P_{XZ|Y}(X, Z|Y)}{P_{X|Y}(X|Y)P_{Z|Y}(Z|Y)} \right] = \mathbb{E} \left[\log_2 \frac{P_{X|Y}(X|Y)P_{Z|Y}(Z|Y)}{P_{X|Y}(X|Y)P_{Z|Y}(Z|Y)} \right] = \mathbb{E}[\log_2(1)] = 0. \quad (9.18)$$

However, we can ask ourselves what is the information between the input X and the other two random variables Y and Z , namely the mutual information $I(X; Y, Z)$ which can be also written as $I(Y, Z; X) = H(Y, Z) - H(Y, Z|X)$ in two ways using the chain rule of entropy. On the one hand, using $H(Y, Z) = H(Y) + H(Z|Y)$ and $H(Y, Z|X) = H(Y|X) + H(Z|X, Y)$ and arranging terms we obtain

$$I(Y, Z; X) = H(Y, Z) - H(Y, Z|X) = \underbrace{H(Y) - H(Y|X)}_{I(X; Y)} + \underbrace{H(Z|Y) - H(Z|X, Y)}_{I(X; Z|Y)} \quad (9.19)$$

that is the **chain rule of mutual information**. As opposed to entropy, conditioning can increase mutual information! On the other hand, using the chain rule of entropy in the second way as $H(Y, Z) = H(Z) + H(Y|Z)$ and $H(Y, Z|X) = H(Z|X) + H(Y|X, Z)$, we can also write

$$I(Y, Z; X) = H(Y, Z) - H(Y, Z|X) = \underbrace{H(Z) - H(Z|X)}_{I(X; Z)} + \underbrace{H(Y|Z) - H(Y|X, Z)}_{I(X; Y|Z)}. \quad (9.20)$$

Combining (9.19) and (9.20) into a single system of equations we obtain that

$$I(X; YZ) = I(X; Y) + \overbrace{I(X; Z|Y)}^{=0} \quad (9.21)$$

$$= I(X; Z) + \underbrace{I(X; Y|Z)}_{\geq 0} \quad (9.22)$$

where $I(X; Z|Y) = 0$ from the Markov chain assumption and $I(X; Y|Z) \geq 0$ since it is a mutual information. Thus, (9.21) and (9.22) lead to the **data processing theorem**: if the random variables $X \rightarrow Y \rightarrow Z$ form a Markov chain, then

$$I(X; Z) \leq I(X; Y). \quad (9.23)$$

The result in (9.23) is a powerful inequality that says that any processing done to a given variable cannot increase the mutual information, or that the information that the output of the chain Z has about X cannot be bigger than the information that Y has about X .

We end the last lecture combining some of the previous results to study the mutual information between $(X_1, \dots, X_n)$, a sequence of random variables that is the input of a **discrete memoryless channel**, and the channel output sequence $(Y_1, \dots, Y_n)$, namely $I(X_1 \cdots X_n; Y_1 \cdots Y_n)$. To do so, we write the mutual information as a function of entropy

$$I(X_1 \cdots X_n; Y_1 \cdots Y_n) = H(Y_1 \cdots Y_n) - H(Y_1 \cdots Y_n | X_1 \cdots X_n) \quad (9.24)$$

$$= H(Y_1 \cdots Y_n) - \sum_{i=1}^n H(Y_i | X_1 \cdots X_n Y_1 \cdots Y_{i-1}) \quad (9.25)$$

$$= H(Y_1 \cdots Y_n) - \sum_{i=1}^n H(Y_i | X_i) \quad (9.26)$$

$$\leq \sum_{i=1}^n H(Y_i) - \sum_{i=1}^n H(Y_i | X_i) \quad (9.27)$$

$$= \sum_{i=1}^n I(X_i; Y_i) \quad (9.28)$$

where (9.25) follows from the chain rule of entropy, (9.26) follows from the fact that the channel is memoryless and thus, the output Y_i only depends on the input X_i , (9.27) follows from (9.8) and the last step from the definition of mutual information. The mutual information between the input sequence and the output sequence in a discrete memoryless channel becomes the sum of symbol-wise mutual informations with equality in step (9.27) only when $(Y_1, \dots, Y_n)$ becomes a sequence of independent random variables, suggesting that this is the best strategy for data transmission over memoryless channels.